ASSIGNMENT 02 · CO3 + CO4

Reconstruct or Create?

A comparative study of four Generative AI models

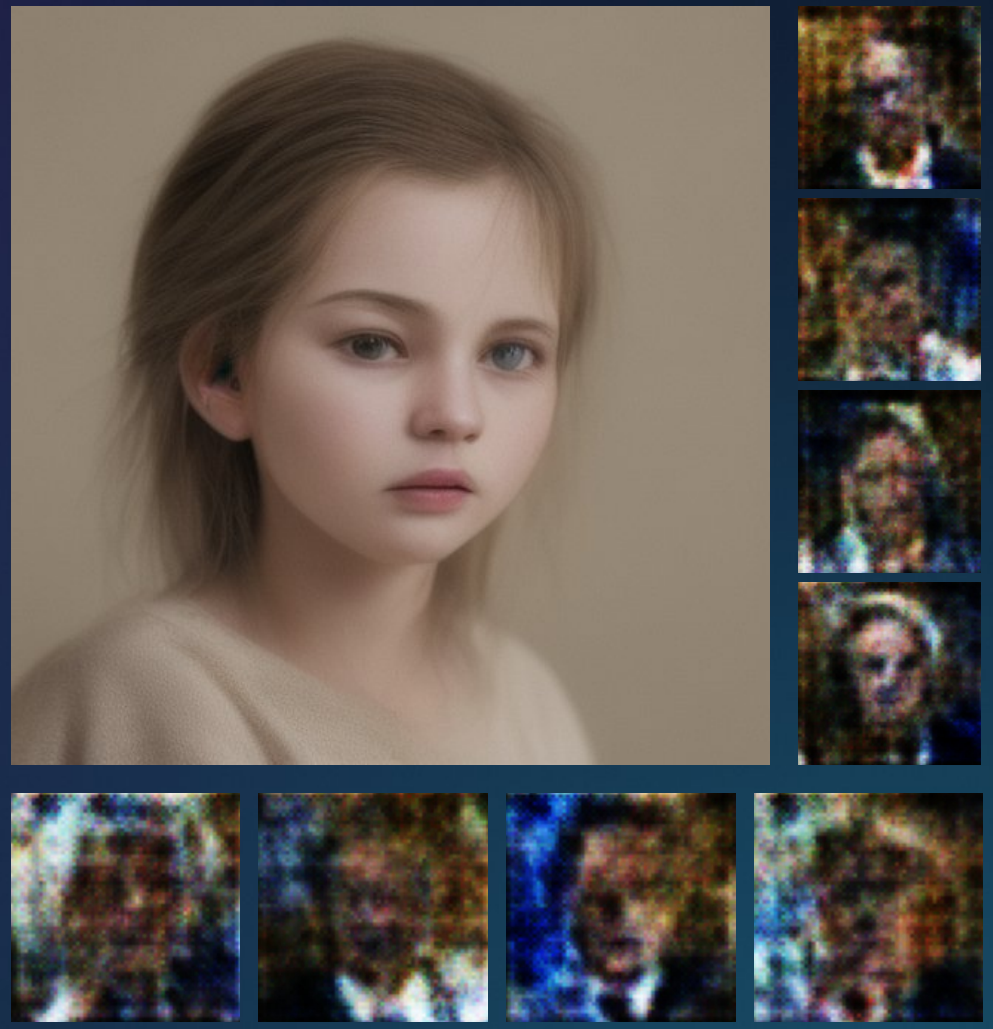
Autoencoder · VAE · DCGAN · Diffusion

Generative AI · B.Tech AIML · MIT Academy of Engineering, Pune

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Open our Colab notebook →



All faces on this slide are AI-generated (diffusion + our DCGAN)

Four models, two jobs

RECONSTRUCTION · CO3

GENERATION · CO4

Autoencoder

Squeezes the image into 128 numbers, then rebuilds it (MSE loss)

- ⊕ Simple, fast, stable
- ⊖ Blurry, loses detail

Used for Compression, denoising

VAE

Learns a distribution (μ, σ), samples z = μ + σ·ε (recon + KL loss)

- ⊕ Can sample new faces
- ⊖ Even softer outputs

Used for Representation learning

DCGAN

Generator fakes faces from noise; discriminator judges real vs fake

- ⊕ New, sharper images
- ⊖ Unstable, artefacts

Used for Face synthesis, augmentation

Diffusion

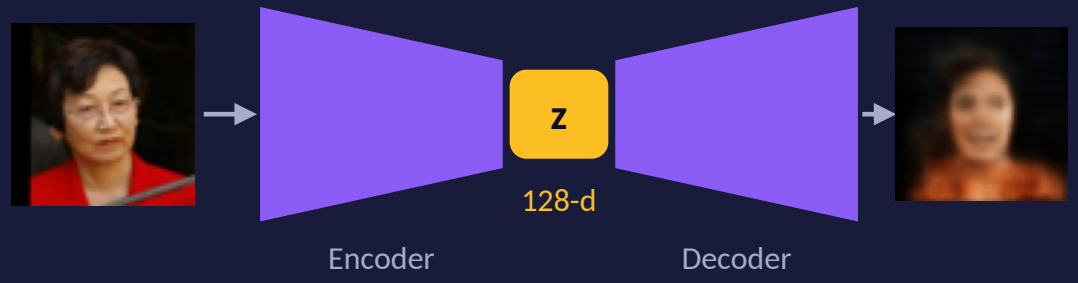
Removes noise step by step, guided by a text prompt (U-Net)

- ⊕ Best quality + prompt control
- ⊖ Heavy compute

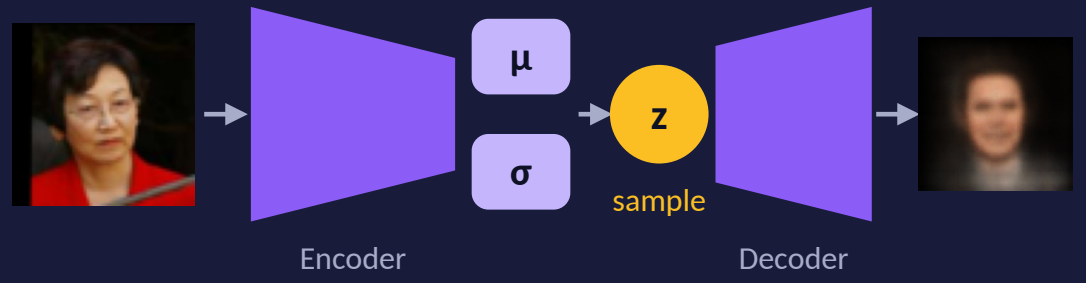
Used for Text-to-image, design

Inside each model

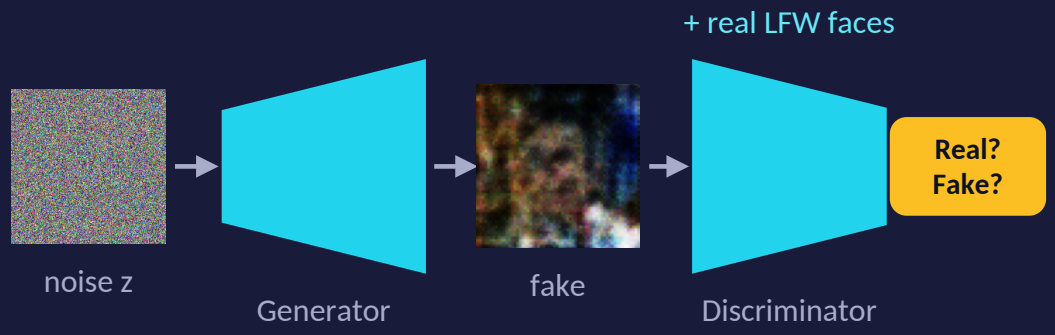
Autoencoder compress → rebuild



VAE learn a distribution, then sample

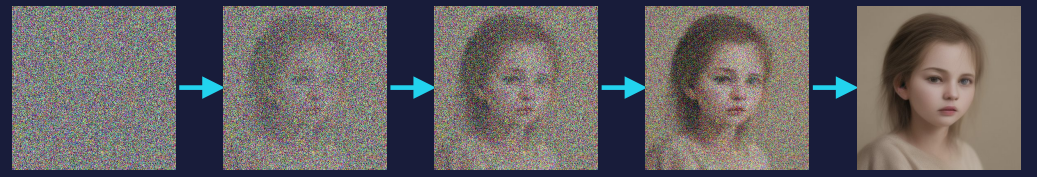


DCGAN generator vs discriminator



Diffusion denoise step by step, guided by text

prompt: "realistic portrait..." · guidance 7.0

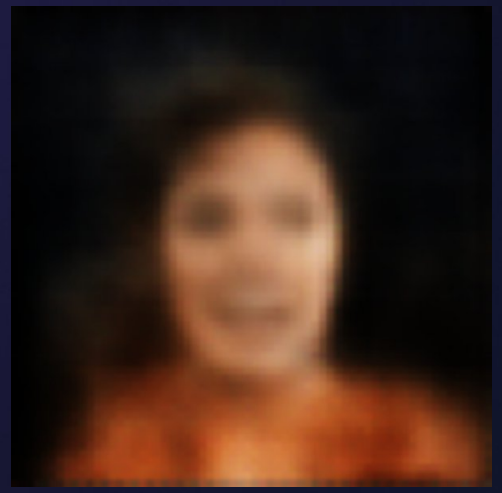


Faces in the AE/VAE diagrams are a real test image and its actual reconstructions. The diffusion strip illustrates noise levels using our generated output.

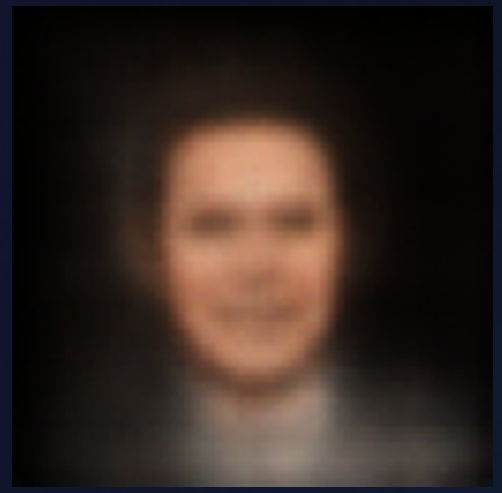
Activity A: AE keeps more detail than VAE



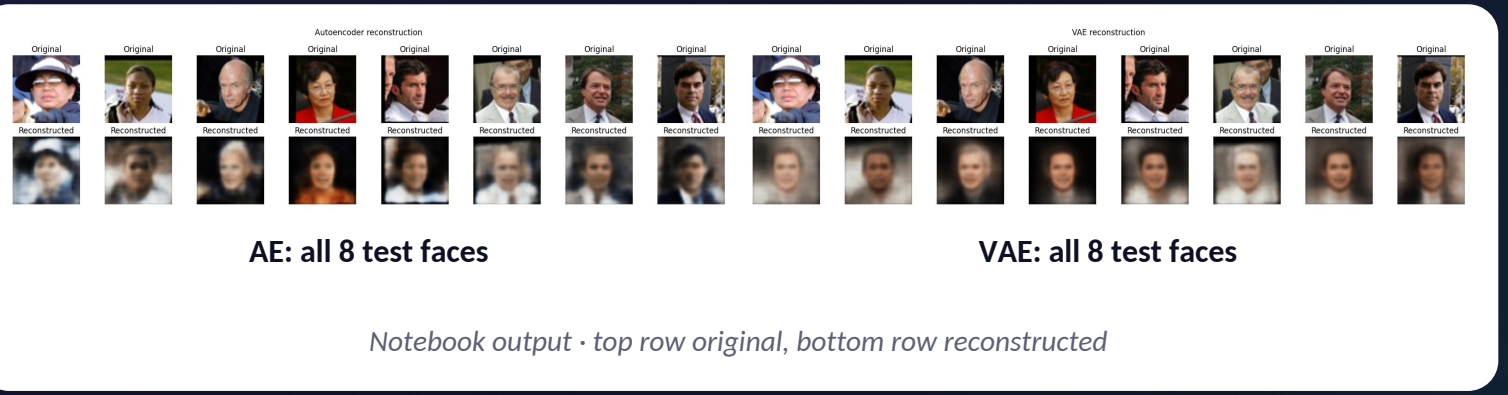
Original (zoomed)



Autoencoder



VAE



AE: all 8 test faces

VAE: all 8 test faces

Notebook output · top row original, bottom row reconstructed

0.0144

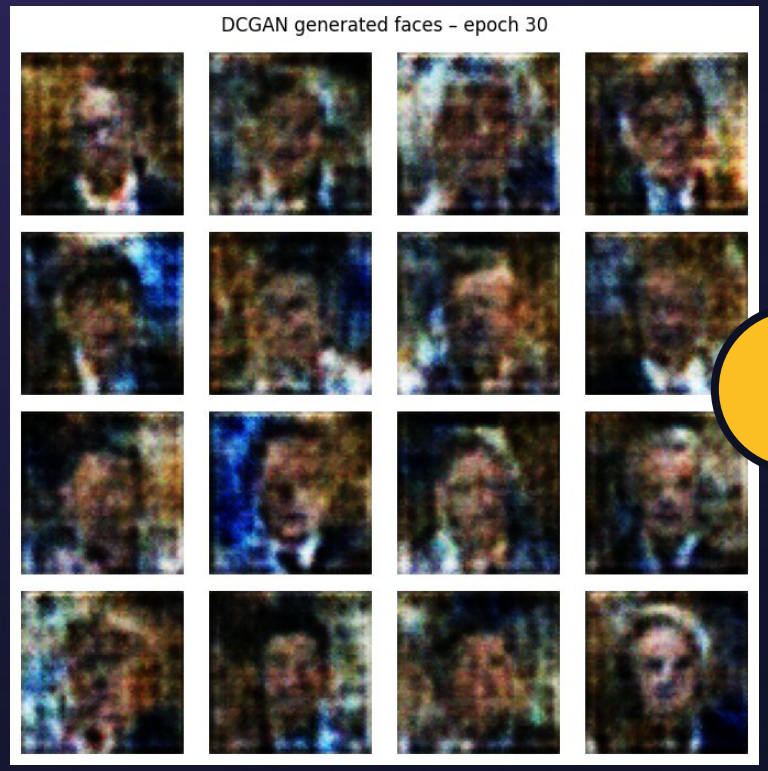
AE validation MSE

−74%

MSE drop in 10 epochs

- AE keeps face shape, skin tone and lighting
- Hair, eyes and edges get blurred
- VAE faces look smoother and "averaged"
- Main difference: loss of detail, not distortion

Activity B: Pretrained Diffusion Produces More Realistic Images in Our Experiment



VS



DCGAN · TRAINED BY US

30 epochs · 64×64 · G 0.79 / D 1.38

DIFFUSION · PRETRAINED

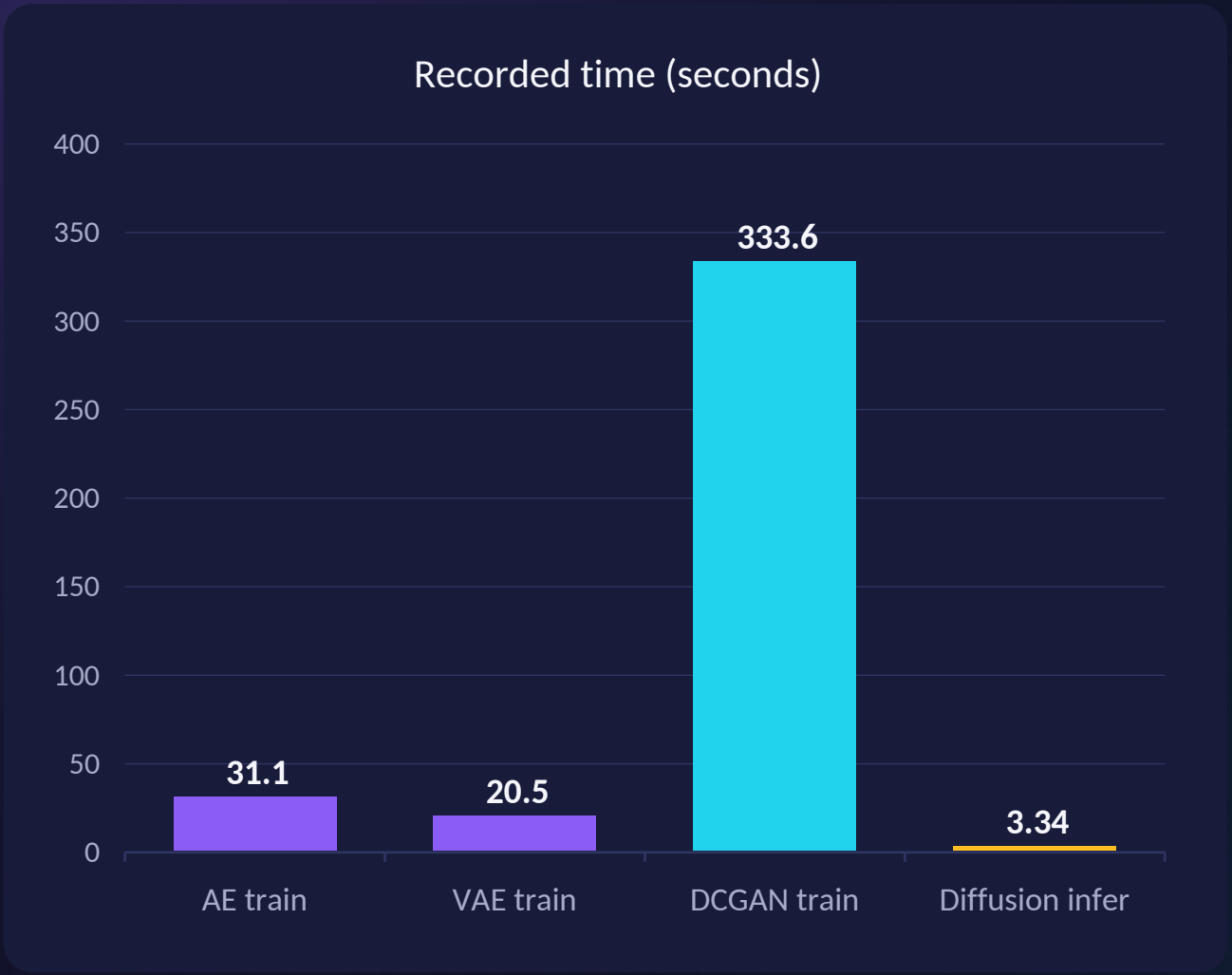
segmind/tiny-sd · 20 steps · guidance 7.0

What we saw

- Generator creates new faces from random noise.
- Some generated images contain colour noise and visual distortions.
- Discriminator loss was approximately 1.38, indicating a near-balanced adversarial loss pattern.
- Image quality was evaluated by visually inspecting the generated samples.

Prompt: "A realistic portrait photograph of a person, soft natural daylight, neutral background, high detail."

DCGAN costs the most; losses are not comparable



10×+

DCGAN vs AE/VAE time

2 nets

trained every step

Model	Loss and scale	Final
AE	MSE, averaged per pixel	0.0144
VAE	BCE + KL, summed	6,772.4
DCGAN	Adversarial BCE	G 0.79 / D 1.38
Diffusion	Pretrained	—

Different loss functions, different scales: compare the images, not the numbers.

What we learned and how we used AI

AI usage and verification

- Claude for concepts, code explanation, the comparison table and the report
- Every number checked against the notebook output
- Theory cross-checked with original papers and docs

Key learning

- AE vs VAE: Reconstruction detail vs organised latent space.
- GAN: Loss values alone do not indicate image quality.
- Diffusion: High-quality, prompt-guided image generation.
- Important: Training time and inference time are different.

Contributions

- Sanika ,Tanishka : AE and VAE experiments
- Tanishka , Amruta : DCGAN and diffusion experiments
- Amruta ,Sanika: comparison, AI documentation, report and PPT

Conclusion AE/VAE for reconstruction · DCGAN for dataset-specific generation · Diffusion for the best quality at the highest cost